



Fig. 3: Relative humidity vs output voltage



Fig. 4: Humidity sensor



Fig. 5: MQ135 air-quality sensor

relative humidity directly into equivalent mV output. The driver circuit is in-built in the sensor assembly (see Fig. 4). The rated voltage is 5V DC. The sensor can measure humidity level from 20 per cent to 90 per cent RH, which are usually the general conditions. Fig. 3 shows the mV output reading corresponding to relative humidity. The output of humidity sensor SENSI is connected to analogue input A4.

Light sensor (NORP12). We have used one of the cheapest light sensors, i.e., an LDR, which gives us a rough estimate of the ambient light intensity. The sensor arrangement is adjusted as per properties of the

particular model. If we use any other LDR, then the sensor arrangement and program need to be changed. The LDR's resistance changes with the amount of light intensity falling on its exposed surface. Light intensities of some common areas are mentioned below for reference:

1. Outdoors on a sunny day > 10,000 Lux
2. Standard living room = 70 to 100 Lux
3. 60W bulb at 1m distance = 50 to 60 Lux

The output from the LDR arrangement is connected to analogue input pin A3.

Air-quality sensor (MQ135). Sensitive material of the sensor is tin

dioxide (SnO_2), whose conductivity increases with the concentration of the gas. The change in conductivity is converted into the output voltage signal, which varies corresponding to the concentration of the combustible gas. MQ135 is highly sensitive to ammonia, sulfide and benzene steam, smoke and other harmful gases. It is a low-cost sensor, suitable for different applications (see Fig. 5). The output of the gas sensor is connected to analogue input A2 of the Arduino board.

Serial LCD. LCD1 used in this project is a serial LCD that saves pins of Arduino for future expansion of this project in terms of alarms, LEDs, etc. It has three pins, viz. Vcc, GND and Rx. The text to be displayed on the screen is sent serially via UART port of Arduino to Rx pin of serial LCD as shown in Fig. 2.

The 12V power supply through the DC jack is regulated to 5 V using IC1 and applied to the circuit.

When the system is powered on, the device takes one minute to preheat the gas sensor before it is ready for operation. The LCD displays the ap-



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